

Resolution Geometry of Operational Duration

Cross-Clock Path Dependence, Rejection of Universal Clock Accumulation, and Conditional Proper-Time Realization

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September 20, 2026

Abstract

Relativistic clock experiments are usually read as evidence that “time itself” flows differently for different observers. This paper separates that operational statement from the geometric formalism used to represent it, and states exactly the version that survives hostile attack. Beginning from locally realized clock mechanisms, we build operational duration without assuming spacetime geometry: three or more independent clock families define a common affine duration object when frozen cross-calibrations satisfy held-out prediction *and* cocycle closure. Controlled path-separation then tests a path-independent universal clock-accumulation hypothesis H_{PI} . The central result is operational, not geometric: identically prepared clocks do not accumulate one path-independent elapsed duration on the tested domain, stated through an affine-gauge-invariant elapsed contrast so that no origin or unit is smuggled in. Crucially, and against an earlier overclaim, this does *not* prove that no universal scalar coordinate exists: we exhibit the counterexample $dT/dt = r(v, U)$, a global coordinate carrying path-dependent clock rates, which reproduces unequal reunited readings exactly. The theorem therefore rejects *path-independent universal clock accumulation*, not the mathematical existence of a coordinate—a distinction between an invariant accumulated along a history and a labeling object. Only afterward is a Lorentzian candidate introduced: proper time is accepted as a realization of the surviving operational duration only when one frozen affine calibration predicts held-out clock families and trajectory classes better than declared competing path laws, and a flexible non-metric competitor shows this is a conditional model identification, not a derivation. Two gates are mandatory—species-by-path exclusion and synchronization/orientation control—and orientation and absolute scale remain outside what clock comparison alone fixes. All claims are checked by an adversarial battery; the ladder separates operational, geometric, and metaphysical statements, and no step derives general relativity from quantum mechanics.

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1 Scope and organizing claim

The paper hangs on one ordering: *clock disagreement is experimentally prior to geometric representation*. The logical order is

clock records $\rightarrow$ operational duration $\rightarrow$ path dependence $\rightarrow$ geometric realization,

not the textbook metric $\rightarrow$ proper time $\rightarrow$ clock behaviour. Each arrow is a separate, attackable step; none is assumed by the next.

2 Operational clock processes

Definition 1 (Clock process). A clock process is a reproducible physical mechanism producing an ordered sequence of observable events, phase accumulations, or count observables.

Definition 2 (Operational duration). For a phase process $\Delta\phi_i$ with declared local transition frequency ω_i , the operational duration is $T_i = \Delta\phi_i/\omega_i$, subject to unwrapping, calibration, and a declared uncertainty model.

Remark 1. No spacetime metric, proper time, Lorentzian structure, or relativity principle is assumed at this layer.

3 Cross-clock affine duration object

Let Q_A, Q_B, Q_C be independent clock families with positive-affine transitions $Q_j = a_{ji}Q_i + b_{ji}$, $a_{ji} > 0$.

Theorem 1 (Affine clock realization). *Suppose frozen affine maps fitted on a calibration domain (i) predict held-out readings within the declared uncertainty, (ii) contain identity and inverses, and (iii) satisfy triple-overlap cocycle closure $f_{ki} = f_{kj} \circ f_{ji}$, i.e. $a_{ki} = a_{kj}a_{ji}$ and $b_{ki} = a_{kj}b_{ji} + b_{kj}$. Then the clock families are charts of one common affine duration object $\mathcal{T}$, unique up to $\mathcal{T} \mapsto a\mathcal{T} + b$, $a > 0$.*

Remark 2 (The third clock is anti-circularity). Pairwise agreement is insufficient; cocycle closure is what certifies that all comparisons are representations of *one* object rather than unrelated pairwise regressions. (Verified: held-out RMSE at the injected noise scale $\sim 10^{-3}$; cocycle parameter error $\sim 5 \times 10^{-8}$.)

4 Controlled path separation and the path-independent null

Identically prepared clocks begin co-located, are transported along γ_A, γ_B , and reunited, with elapsed durations $\Delta T_A, \Delta T_B$.

Definition 3 (Path-independent universal accumulation, H_{PI}). There is one common law G with $\Delta T[\gamma] = G(\Delta t)$ depending only on the elapsed coordinate interval; equivalently, identical ideal clocks satisfy $\Delta T_A = \Delta T_B$ whenever $\Delta t_A = \Delta t_B$.

5 Gauge-invariant elapsed-duration contrast

Because $\mathcal{T} \mapsto a\mathcal{T} + b$ is gauge freedom, the falsification statistic must not depend on an origin or unit. Use

$$D_{AB} = \Delta T_A - a_{AB}^{-1} \Delta T_B, \quad Z_{AB} = D_{AB} / \sigma(D_{AB}).$$

Proposition 1. *Whether $D_{AB} = 0$, the sign of a nonzero D_{AB} , and normalized ratios are invariant under positive affine gauge changes.* (Verified to floating-point precision; sign preserved under $\mathcal{T} \mapsto 3\mathcal{T} + 5$ and $\mathcal{T} \mapsto 0.2\mathcal{T} - 1$.)

6 Main theorem, correctly narrowed

Theorem 2 (Cross-clock path-dependent duration). *Assume: three or more independent clock families define a held-out, cocycle-consistent affine clock object; identically prepared clocks follow distinct controlled histories; the affine-gauge-invariant contrast satisfies $D_{AB} \neq 0$ with declared significance; and synchronization error, transport offsets, environmental biases, and species-by-path interactions are excluded within the declared uncertainty model. Then H_{PI} is rejected on the tested domain:*

identical ideal clocks do not accumulate one path-independent universal elapsed duration.

(Synthetic witness: $Z_{AB} \approx 23$.)

7 The universal-coordinate loophole: the essential non-claim

Non-claim 1 (No proof that a coordinate cannot exist). The theorem does *not* establish $\nexists t$. A global scalar coordinate t may exist while clocks obey a path-dependent local rate

$$\frac{dT}{dt} = r(v, U, \dots), \quad T_A = \int r(v_A, U_A) dt, \quad T_B = \int r(v_B, U_B) dt,$$

so that $\Delta T_A \neq \Delta T_B$ even though both are parameterized by the same t .

This is the decisive narrowing forced by attack. (Verified: with $dT/dt = 41\sqrt{1-v^2}(1+U)$ and a single shared coordinate interval, two paths give $T_A = 44.19 \neq T_B = 43.72$, reconstructed from the common coordinate to error $\sim 10^{-14}$.) A coordinate is a labeling object; a clock duration is an invariant accumulated along a history. Their types differ, and general relativity itself permits coordinate times while denying that coordinate time is the universal proper duration recorded by every clock. The paper therefore attacks *absolute universal duration*, not the bare existence of a scalar coordinate; without this narrowing the loophole breaks the result.

8 Mandatory gates

Proposition 2 (Species-by-path gate). *Path dependence must be shown shared across clock families, not species-specific device response. After the frozen affine map, the residual $\varepsilon_{BA}(\gamma) = Q_B(\gamma) - (a_{BA}Q_A(\gamma) + b_{BA})$ must satisfy $\text{Corr}(\varepsilon_{BA}, X) \approx 0$ for all declared path variables $X \in \{v, U, a, T_{\text{env}}, \dots\}$. (Verified negative control: an injected v^2 sensitivity produces residual correlation ≈ 1.0 , which the gate flags as drift, not shared duration.)*

Proposition 3 (Orientation and synchronization gates). *Unsigned elapsed duration is invariant under path reversal ($\tau[\gamma^{-1}] = \tau[\gamma]$), so it does not fix time orientation, a preferred foliation, or a global time function. Reunion differences are contaminated by synchronization offsets $\delta_A - \delta_B$; the hypotheses must include co-location before separation, common departure/reunion comparison, reversal/swap controls, label randomization, and independent synchronization channels.*

9 Conditional proper-time realization

Introduce a frozen Lorentzian candidate class $[g]$ with $\tau_g[\gamma] = \frac{1}{c} \int_\gamma \sqrt{-g_{\mu\nu} dx^\mu dx^\nu}$ and the realization hypothesis $T[\gamma] = a \tau_g[\gamma] + b$.

Theorem 3 (Conditional proper-time realization). *Suppose the affine clock object is established; path rank is sufficient; gauge and signature are fixed and stable; the Lorentzian model is frozen before testing; one global affine calibration (a, b) predicts all held-out clock species and trajectory classes; and declared competing path laws perform worse. Then τ_g is supported as an empirical realization of the common operational duration on the tested domain.*

Remark 3 (Realization is identification, not derivation). Good proper-time prediction does not uniquely select Lorentzian geometry: a sufficiently flexible path-dependent regression can reproduce much of the structure. (Verified: held-out RMSE $\sim 7 \times 10^{-4}$ for frozen τ_g versus 6.24 for the path-independent null; a flexible non-metric path model sits between.) The realization is therefore a non-faithful model identification against declared competitors, not a derivation of a unique latent spacetime, and in particular *not* a derivation of general relativity from quantum mechanics.

10 Einstein, Minkowski, and proper time

The operational ladder clock records $\rightarrow$ operational duration $\rightarrow$ path dependence $\rightarrow$ proper-time realization is logically distinct from metric $\rightarrow$ proper time $\rightarrow$ clock behaviour. Read historically: Einstein (1905) is the operational statement $T[\gamma_A] \neq T[\gamma_B]$; Minkowski (1908) is the geometric representation $c^2 d\tau^2 = c^2 dt^2 - d\mathbf{x}^2$; general relativity (1915) is the dynamic metric $d\tau^2 = -c^{-2} g_{\mu\nu} dx^\mu dx^\nu$.

Corollary 1. *Relativistic clock disagreement need not be read first as deformation of a pre-existing universal time. It is established first as path dependence of operational duration; Lorentzian proper time is then tested as a geometric realization.*

11 The locked K0–K7 ladder

Rung	Object	Status
K0	local clock observable $T_i = \Delta\phi_i/\omega_i$	declared operational representation
K1	three independent clock families, co-located calibration	experimental calibration gate
K2	frozen positive-affine maps, held-out + cocycle closure	conditional affine-clock theorem
K3	controlled path separation γ_A, γ_B	declared experiment
K4	reject H_{PI} via gauge-invariant D_{AB}	empirically falsifiable
K5	species-by-path exclusion	systematic-exclusion gate
K6	frozen Lorentzian model vs declared competitors	conditional geometric realization
K7	held-out cross-clock, multi-path proper-time prediction	conditional empirical identification

Theorem 4 (Locked statement). *On a tested domain, if three or more independent clock families define a held-out, cocycle-consistent affine clock; identically prepared clocks along distinct controlled histories show reproducible affine-gauge-invariant elapsed differences; species-by-path and synchronization explanations are excluded within a declared complete uncertainty model; and a frozen Lorentzian proper-time functional with one global affine calibration predicts all held-out clocks and paths better than declared alternatives—then the experiment rejects path-independent universal clock accumulation on that domain and supports Lorentzian proper time as an empirical representation of the common operational duration. It does not exclude the mathematical existence of an auxiliary global coordinate carrying path-dependent clock-rate laws.*

12 Adversarial battery and reproducibility

The analytic results do not depend on simulation; the battery seeks counterexamples and calibrates the gates. Under seed 20260920: three-clock held-out affine calibration survives (RMSE $\sim 10^{-3}$); cocycle closure survives (error $\sim 10^{-8}$); the restricted absolute-clock null is falsified in the synthetic witness ($Z \approx 23$); the universal-coordinate loophole breaks the strong theorem (reconstruction error $\sim 10^{-14}$); the species-by-path gate fires on injected v^2 sensitivity (residual correlation ≈ 1); affine-gauge invariance of the contrast holds to floating point; orientation is undetermined by unsigned duration; and the frozen Lorentzian model beats the path-independent null on held-out paths while a flexible competitor remains partially adequate (conditional realization). All numbers are reproducibility metadata, not premises of the theorems.

13 Non-claims

Non-claim 2. No proof that a universal scalar coordinate cannot exist. A global coordinate may coexist with path-dependent clock-rate laws; the theorem rejects only path-independent universal elapsed-duration accumulation.

Non-claim 3. No derivation of general relativity from quantum mechanics. Proper time is a conditional realization against declared competitors, non-faithful and domain-limited.

Non-claim 4. Orientation and absolute scale are not fixed by clock comparison: unsigned duration is reversal-invariant, and one external calibration constant is required for units.

Non-claim 5. Path dependence is established only where the species-by-path and synchronization gates are closed within the declared uncertainty model.

14 Conclusion

The strongest form that survives every executed attack is not “quantum clocks prove there is no universal time.” It is three separable achievements: independent clocks construct one affine operational-duration object; identical clocks falsify path-independent universal accumulation on the tested domain; and, only afterward, one held-out cross-clock, multi-path calibration can realize the surviving duration as Lorentzian proper time against declared competitors. The essential correction is the replacement of “no universal scalar coordinate exists” by “no path-independent universal clock accumulation represents measured elapsed duration.” With the three-clock cocycle, the species-by-path gate, the affine-invariant contrast, the frozen-competitor comparison, and held-out proper-time prediction, this is a defensible K0–K7 ladder that keeps operational fact, geometric model, and metaphysical coordinate claims cleanly apart.

A Affine clock category

Objects (Q_i, R_i) ; morphisms $f_{ji}(x) = a_{ji}x + b_{ji}$, $a_{ji} > 0$. Identity, inverse $f_{ij} = f_{ji}^{-1}$, and cocycle closure $f_{ki} = f_{kj} \circ f_{ji}$ make **AffClock** a groupoid; its one-object skeleton is the common affine duration object $\mathcal{T}$.

B Universal-coordinate loophole counterexample

With $dT/dt = a\sqrt{1-v^2}(1+U)$ and a single coordinate t , $T_A = \int r(v_A, U_A) dt \neq T_B = \int r(v_B, U_B) dt$ while t exists and reconstructs both readings exactly. This blocks any inference from $\Delta T_A \neq \Delta T_B$ to $\nexists t$.

C Species-by-path attack

Residual $\varepsilon_{BA} = Q_B - (a_{BA}Q_A + b_{BA})$ must satisfy $\text{Corr}(\varepsilon_{BA}, X) \approx 0$ for all declared path variables; an injected $Q_i^{\text{obs}} = Q_i + \eta_i v^2$ produces $\text{Corr}(\varepsilon, v^2) \approx 1$ and is rejected.

D Adversarial batteries

Affine cocycle attack; path-independent null attack; universal-coordinate loophole attack; species/path attack; held-out competitor comparison; synchronization attack; orientation no-go; affine-gauge invariance attack. Seeds and ledgers accompany the executed scripts.

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